# Replication of the orthogonal latent-mean contrasts in Jeffries (2024)

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**Abstract**

Do the latent-mean contrasts in Jeffries (2024) sit on the printed scale −0.40, −0.20, 0, and do the rejection rates in Tables 4.1 and 4.3 follow from that model? Jeffries defines three codes for four groups, states that imbalance makes the set non-orthogonal, and reports multigroup and MIMIC averages near those three numbers when the generating means are $1, 0.6, 0.8, 0.4$. The fitting was done in lavaan 0.6.17. Population contrasts were obtained by fitting each candidate identification to the population moments. A 400-replicate Monte Carlo then used the analysis string from the public OSF project. On a population in which every loading is 0.70, a marker loading of 1 produces −0.28, −0.14, and 0. On the population lavaan builds from the generation string, the first loading is 1 and the other three are 0.70, and the same marker model produces −0.40, −0.20, and 0. The second population matches the printed averages. At $N = 240$, a 400-replicate screen put six of 30 rejection rates outside an uncorrected two-standard-error band, and one of those six outside Bonferroni and Holm: 2.26 against the printed 5.18. The same ten cells at 10,000 replicates put that rate at 5.01. Three rates remain outside the uncorrected band. None remain outside Bonferroni or Holm. A binomial count of those three misses does not exceed the 5% expectation. Redrawing the three cells with simsem 0.5.16 and the `exportData` seed puts two of them inside that band and leaves one outside: 6.46 against 7.41. The dissertation's percents are single Wald tests at $\alpha = 0.05$. The correction applies only to this agreement check. The unbalanced power rates come out 49.35, 21.28, and 8.51 against 49.23, 21.08, and 7.77. Multigroup convergence in the smallest groups is 89.82% against 89.29%.

## 1 What this replication did

This section answers what was fit, and in which program. lavaan 0.6.17 is the estimator named in the dissertation, and it is the program that produced every fitted number below. simsem 0.5.17, the installed successor of the named simsem 0.5.16, was used to parse the generation string. OpenMx and MplusAutomation are not installed. Mplus is not installed. The file `simulation/mgcfa_osf.inp` mirrors the multigroup string and was not executed. `exportData(program = "Mplus")` in the generation scripts writes data files. The estimation scripts call `lavaan::sem`.

Lean is used only for the code identities already proved in `LatentContrasts/Orthogonality.lean`: the products of the Table 2.2 codes, the population values of equations (3.3)–(3.5), and the treatment coding of those codes. The gap between −0.28 and −0.40 is a question about which population and which identification lavaan is fitting. That question is answered by the moment fits, not by a new proof.

The moment fits are `simulation/candidate_readings.R`. The short Monte Carlo is `simulation/validate_gaps.R`, 400 replicates, seed 13131 under R's L'Ecuyer generator, with one drawn seed per cell. The 10,000-replicate rerun is `simulation/validate_10k.R`. Each of its ten

cells has a fixed seed, set once, and the replicates are drawn in order on one core. The agreement classification, including the Bonferroni and Holm checks, is `simulation/score_gaps.R`. Package versions are in `simulation/out/session_info.txt`: R 4.3.3, lavaan 0.6.17, simsem 0.5.17. The miss-cell redraw is `simulation/env_miss.R`, simsem 0.5.16, seed 13131. The dissertation names R 4.2.2.

Jeffries studies one factor with four indicators and four groups. The true-difference means are $\alpha = (1, 0.6, 0.8, 0.4)$. The null means are all 1. Total sample size is 240, 480, or 960. The group shares are 25/25/25/25, 35/35/15/15, and 45/45/5/5.

## 2 Comparison

Table 1 puts one published claim on each row. "Match" means the printed value and our value agree as exact numbers, as the same population target, or, for a Monte Carlo rate, as a gap no larger than two combined standard errors. That rate rule is one comparison at a time. The combined standard error uses the printed count of 10,000 and our count of converged replicates. "Partial" means one piece holds and another does not. "No match" means the reading is on a different scale. The last two rows are the agreement check on all 30 rates, uncorrected and then corrected. The dissertation's own $\alpha = 0.05$ columns are single tests. They are not re-thresholded here.

| Claim | Published | Ours | Verdict |
|---|---|---|---|
| Unweighted products of the Table 2.2 codes | Equation (2.4) is zero | Proved zero | match |
| Contrasts on $(1, 3/5, 4/5, 2/5)$ | $-0.4,\ -0.2,\ 0$ | $-2/5,\ -1/5,\ 0$ | match |
| Third contrast while the means differ | Used as a null contrast | Proved equal to 0 | match |
| Imbalance destroys orthogonality | The set becomes non-orthogonal | Only code 1 with code 3 fails | partial |
| OSF marker model, generation string as lavaan parses it | Averages near $-0.40,\ -0.20,\ 0$ | Population fit $-0.40,\ -0.20,\ 0$ | match |
| Every loading 0.70, then marker loading 1 | The same averages | Population fit $-0.28,\ -0.14,\ 0$ | no match |
| Table 4.1, $N = 240$, shares 45/45/5/5, percent significant | 49.23, 21.08, 7.77 | 49.35, 21.28, 8.51 | match |
| Table 4.3, same cell, percent significant | 8.13, 8.59, 7.73 | 8.10, 8.33, 8.22 | match |
| Table 4.5, same cell, multigroup convergence | 89.29% | 89.82% of 10,000 | match |
| Thirty rejection rates, two combined standard errors | Each printed percent is one test at $\alpha = 0.05$ | 3 of 30 outside the band | partial |
| Same 30 rates, Bonferroni $z = 2.94$ and Holm | Those tests are a single-test design | 0 of 30 outside the band | match |

Table 1: Published claims against this replication. Monte Carlo percent rows use 10,000 replicates at $N = 240$. The population fits use the moment matrices. The last two rows correct only the agreement check. The 400-replicate screen is in Section 3.

The product row is equation (2.4), proved at 103. The population row is equations (3.3)–(3.5) on the printed means, proved at 123. The third contrast is zero while the first two means differ, proved at 127. The imbalance row is partial because 68 shows that sizes $(a, a, b, b)$ leave two of the three weighted products at zero and set the other to $2(b - a)$. At $N = 240$ that product is $-96$ in

condition 2 and −192 in condition 3.

The two scale rows are the gap that used to read as a no match for the whole table. They are different populations. Section 3 scores every reading against the same printed target. The percent rows use the population that matches −0.40. The old comparison of 7.77 with 4.5 was the every-loading-0.70 marker run, whose $z$ statistics sit on the rescaled contrasts. That run is not the row in Table 1.

## 3 Where the gaps come from

The question is which population and which identification produce −0.40, −0.20, and 0, and whether that same fit reproduces the printed rejection rates.

The generation string in the public OSF script, and the repaired form of the appendix string, is

```
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~ 1*1
eta ~ c(1, .6, .8, .4)*1
```

with residual variances 0.51. lavaan 0.6.17 keeps a free loading on `i1` and drops the second modifier 0.7. `simulateData` fills the free loading with 1. A draw of 20,000 at factor mean 1 had mean 1.996 on `i1` and 1.693 on `i2`. The targets are 2 and 1.7. The variances were 1.021 and 0.749. The targets are 1.01 and 0.755. simsem 0.5.17 rejects the page's token `i1 ~ 1*` with the same parse error. The OSF generation script already writes `1*1`. The dissertation page writes `1*` for that one intercept. Figure 3 is the page.

The analysis string in `osf/analysis/mgcfa_model.R` is a marker loading of 1, free factor variances, four free means, and one intercept shared by all four indicators and all four groups. It does not constrain the loadings equal across groups, and it does not fix them at 0.7. The summary script sets the printed standard-error column to the standard deviation of the estimates divided by $\sqrt{10{,}000}$.

### 3.1 Parameter ledger

Status is what the source supports. "Printed" is on the page or in the public script. "Inferred" is a consequence of fitting that string in lavaan 0.6.17. "Assumed" is a choice this replay still had to make.

| Parameter | Status | Value and source |
|---|---|---|
| Loadings in the Chapter III sentence | printed | 0.70 for all four indicators. PDF page 89. |
| Factor variance, residual, intercept | printed | 0.50, 0.51, and 1. The 49% sentence is $1 - \lambda^2$ and needs a unit factor variance. The appendix numbers do not. |
| Latent means | printed | $(1, 0.6, 0.8, 0.4)$ and, in the null, all 1. |
| Table 2.2 codes | printed | $C_1 = (-1, 1, -1, 1)$, $C_2 = (-1, -1, 1, 1)$, $C_3 = (1, -1, -1, 1)$. |
| Appendix and OSF code `c3` | printed | The opposite sign. Groups 1 and 4 are $-1$. The population value is 0 either way. |
| Marker sentence | printed | First loading fixed at 1, intercepts equated, latent variances free. PDF page 90. |
| Analysis string | printed | OSF `mgcfa_model.R`. Marker 1, one common intercept, four free means. |
| First loading under the generation string | inferred | Free, and filled with 1. Loadings 2–4 stay at 0.7. `out/appendix_lavaanify.csv`. |
| `i1 ~ 1*` on the PDF page | printed | Does not parse in lavaan 0.6.17 or simsem 0.5.17. |
| `i1 ~ 1*1` in the OSF script | printed | The token this replay generates from. |
| Standard-error column | inferred | `sd(estimate) / sqrt(n)` in the OSF summary script. |
| Significance | printed | Two-sided $p < 0.05$ from the MLR parameter table. |
| Software versions | printed | lavaan 0.6.17, simsem 0.5.16, R 4.2.2, MLR, 10,000 iterations. This replay uses lavaan 0.6.17, simsem 0.5.17, R 4.3.3, and `iter.max` 10,000. |
| Random stream | assumed | L'Ecuyer, `set.seed(13131)`, then one seed per cell. Not the simsem stream. |
| MIMIC latent intercept | inferred | The posted line `eta ~ c1 + c2 + c3` fixes it at 0. Adding an intercept matches the printed averages. |

### 3.2 Each reading against the printed averages

Table 2 scores each reading against the printed target $-0.40$, $-0.20$, 0. Moment rows are the lavaan fit to the population mean vector and covariance, with $10^7$ observations per group, so Monte Carlo error is not the comparison. MIMIC rows are one sample of 100,000 per group, seed 7, except the `contr.sum` row, which is the 20,000-per-group sample from seed 13131.

Two different constraints land on $-0.40$: fixing the measurement at the equal-loading population, and leaving the first loading at the value lavaan generates from `NA*i1`. The public analysis script is the second constraint. It is a marker model. On the equal-loading population that script returns $-0.28$. On the parsed generation string it returns $-0.40$. The printed tables match the second of those. The earlier replay generated every loading at 0.70 and then fixed the first at 1, which is the first row of Table 2. Its Monte Carlo averages, $-0.286$, $-0.134$, and 0.007 at 200 replicates, are that row and not a failure of the optimizer.

A free latent intercept is required for the MIMIC averages. The line in `osf/analysis/mimic_model.R` does not include one, and lavaan 0.6.17 then fixes the latent intercept at 0. The large-sample slopes are about $-0.15$, 0.06, and 0.27. The saved summary is near $-0.40$. The fit that adds the intercept lands on $-0.399$, $-0.200$, and 0.000. The posted loop also assigns the $z$ statistic back onto the estimate before the row is stored. The saved summary contains values near $-0.40$, so that summary was not produced by that assignment.

| Reading | Ours | Explanatory parameter | Verdict |
|---|---|---|---|
| All loadings 0.70, marker, first intercept 0 | −0.280, −0.140, 0 | Marker rescales by 0.7 | no match |
| All loadings 0.70, OSF common-intercept marker | −0.280, −0.140, 0 | Same rescaling | no match |
| All loadings 0.70, loadings and intercepts fixed | −0.400, −0.200, 0 | Measurement fixed at the population | match |
| All loadings 0.70, factor variance fixed at 0.5, first loading free | −0.400, −0.200, 0 | Scale set by $\psi$ | match |
| Parsed generation string, OSF marker | −0.400, −0.200, 0 | First loading generated at 1 | match |
| Parsed generation string, marker, first intercept 0 | −0.400, −0.200, 0 | A common shift of the means | match |
| Parsed string, group-1 mean fixed at 0 | −0.276, −0.115, −0.038 | Reference mean under a shared intercept | no match |
| Parsed string, loadings fixed at 0.7 | −0.431, −0.215, −0.001 | The first loading is not 0.7 | no match |
| Factor variance 1, loadings 0.70, marker | −0.280, −0.140, 0 | Marker still rescales by 0.7 | no match |
| Wishart likelihood versus normal, OSF model | Difference below $10^{-13}$ | $n$ against $n-1$ | no move |
| ML versus MLR, one large sample | Difference 0 | Robust standard errors | no move |
| MIMIC, latent intercept free, factor codes | −0.399, −0.200, 0.000 | Treatment contrast equals the defined contrast | match |
| MIMIC, posted line, intercept fixed at 0 | −0.150, 0.062, 0.273 | The fixed intercept | no match |
| MIMIC, numeric codes ±1, intercept free | −0.208, −0.097, 0.001 | The code changes by 2 | no match |
| MIMIC, numeric effects codes ±1 | Half the contrast, sign follows the code | Code direction | no match |
| MIMIC, `contr.sum` assigned to the factors | Same slope as treatment coding | lavaan did not switch the slope | no move |

Table 2: Candidate readings against the printed averages −0.40, −0.20, 0. The printed column is that target on every row.

### 3.3 Monte Carlo noise

The question is whether the remaining percent gaps are the size of a 400-replicate simulation. A gap is outside noise when it exceeds two combined standard errors. For a percent, the combined standard error is

$$100\sqrt{\frac{\hat{p}(1-\hat{p})}{10{,}000}+\frac{\tilde{p}(1-\tilde{p})}{n}},$$

with $\hat{p}$ the printed rate and $n$ our number of converged replicates. For an average, the printed standard error is already the standard error of the mean of 10,000 replicates, and ours is the standard deviation divided by $\sqrt{n}$.

All 30 averages in the $N = 240$ cells we ran, multigroup and free-intercept MIMIC, true and null, are inside that band. The sampling standard deviations match the printed standard errors times 100. In the multigroup 45/45/5/5 true-difference cell the printed standard errors are all 0.0021, so the implied standard deviations are 0.210. Ours were 0.210, 0.221, and 0.206.

The 7.77 against 4.5 comparison does not survive this population. The 4.5 was the third contrast's rejection rate in the equal-loading marker run. Here the same printed rate is 7.77 and ours is 7.08. The combined standard error is 1.39 points.

| Printed cell, $N = 240$ | Ours | Verdict |
|---|---|---|
| Table 4.1, equal groups, averages $-0.4046$, $-0.2027$, $-0.0010$ | $-0.412$, $-0.204$, 0.004 | within noise |
| Table 4.1, shares 45/45/5/5, averages $-0.4038$, $-0.2181$, 0.0054 | $-0.394$, $-0.222$, 0.011 | within noise |
| Table 4.1, that cell, percent significant 49.23, 21.08, 7.77 | 48.16, 25.21, 7.08 | within noise |
| Table 4.3, that cell, percent significant 8.13, 8.59, 7.73 | 10.80, 12.85, 9.77 | partial |
| Table 4.5, that cell, convergence | 353/400 = 88.25% against 89.29% | within noise |

Table 3: The cells that motivated the gap check. Multigroup model, parsed generation population, 400 attempted replicates.

Six of the 30 rejection rates are outside the uncorrected band. All six are percents. Their averages are inside it. The other 24 rates are inside the same band. $|z|$ is the absolute difference divided by the combined standard error.

| Cell | Contrast | Printed | Ours | $\lvert z\rvert$ | Corrected |
|---|---|---|---|---|---|
| Multigroup, equal, true difference | 3 | 5.29 | 3.27 | 2.19 | inside |
| Multigroup, 35/15, null | 2 | 5.15 | 9.02 | 2.67 | inside |
| Multigroup, 45/5, null | 2 | 8.59 | 12.85 | 2.48 | inside |
| MIMIC, 45/5, true difference | 1 | 62.98 | 57.89 | 2.02 | inside |
| MIMIC, 45/5, true difference | 2 | 23.37 | 18.55 | 2.42 | inside |
| MIMIC, equal, null | 1 | 5.18 | 2.26 | 3.77 | outside |

The corrected column is the same under three rules: Bonferroni at $z = \Phi^{-1}(1-0.05/30) = 2.94$, the two-sided Bonferroni threshold $z = \Phi^{-1}(1-0.05/60) = 3.14$, and Holm. One rate stays outside. It is the MIMIC model, equal groups, null means, contrast 1: 2.26 against 5.18, $z = -3.77$, two-sided $p = 0.00016$, from 399 converged replicates.

### 3.4 Multiplicity

The percent columns in Tables 4.1–4.4 are one Wald test on one dataset at $\alpha = 0.05$. That is the design of those columns. This replication leaves those tests at $\alpha = 0.05$. Multiplicity is a property of the agreement check only: thirty differences between a printed rate and our rate.

The uncorrected check is a two-sided 5% test, $|z| > 1.96$. The two-combined-standard-error band is that rule with 2 in place of 1.96. On these thirty rates the two cutoffs select the same six rows.

Bonferroni divides 0.05 by 30. The one-sided normal quantile of the remaining tail is 2.94. The two-sided quantile is 3.14. Holm sorts the thirty two-sided $p$-values and compares the $i$th with $0.05/(31 - i)$, then stops at the first $p$-value that fails its cutoff. The smallest $p$-value is 0.00016, below $0.05/30 = 0.00167$, so Holm rejects that row. The next $p$-value is 0.0076, from the multigroup 35/15 null contrast 2. Holm's second cutoff is $0.05/29 = 0.0017$. The step-down stops. Five of the six uncorrected misses are inside every corrected rule above.

A binomial test asks whether six misses in thirty trials exceeds the count expected at $p = 0.05$. The upper tail is 0.0033. Under independence the expected count is 1.5. The thirty indicators are dependent. Each cell contributes three contrasts computed on the same replicates, so a miss on one contrast moves with the other two. Dependence of that kind widens the distribution of the miss

count relative to a Bin(30, 0.05) draw. The binomial probability is the independent-trials figure. The ten cells use separate seeds, so the dependence sits inside cells. At 400 replicates the Table 4.3 contrast 2 rate, 12.85 against 8.59, is the uncorrected miss in that cell, and it is inside the corrected band.

The multigroup null model at $N = 240$ and shares 45/45/5/5 converged in 389 of 400 replicates, 97.25%. Table 4.5 prints 9,675 and 95.75% on that row. The count divided by 10,000 is 96.75.

### 3.5 Ten thousand replicates

The question is whether the six uncorrected misses, and the one corrected miss, were the length of the 400-replicate screen. The rerun is `simulation/validate_10k.R`: the same ten cells, 10,000 attempts, `missing = "ML"`, MLR, and `iter.max` of 10,000. Each cell sets one L'Ecuyer seed and draws its replicates in order on one core. The seeds are in `simulation/out/seeds_10k.csv`. The scores are `simulation/out/gap_scores_10k.csv`.

Five of the six short-run misses fall inside the uncorrected band.

| Cell | Contrast | Printed | At 400 | At 10,000 | $\lvert z\rvert$ |
|---|---|---|---|---|---|
| Multigroup, equal, true difference | 3 | 5.29 | 3.27 | 5.23 | 0.19 |
| Multigroup, 35/15, null | 2 | 5.15 | 9.02 | 5.19 | 0.11 |
| Multigroup, 45/5, null | 2 | 8.59 | 12.85 | 8.33 | 0.66 |
| MIMIC, 45/5, true difference | 1 | 62.98 | 57.89 | 62.13 | 1.25 |
| MIMIC, 45/5, true difference | 2 | 23.37 | 18.55 | 22.11 | 2.12 |
| MIMIC, equal, null | 1 | 5.18 | 2.26 | 5.01 | 0.55 |

The row that Bonferroni and Holm had kept outside, 2.26 against 5.18, is 5.01 against 5.18. One of the six stays outside the uncorrected band: MIMIC, shares 45/45/5/5, true difference, contrast 2, 22.11 against 23.37, $|z| = 2.12$. Its average is $-0.2012$ against $-0.2007$.

Across all 30 rates, three miss the uncorrected band. The other two were inside it at 400 replicates. MIMIC, shares 45/45/5/5, null, contrast 2 is 6.46 against 7.41, $|z| = 2.64$. MIMIC, equal groups, null, contrast 3 is 4.53 against 5.32, $|z| = 2.59$. The smallest two-sided $p$-value is 0.0083. Bonferroni's cutoff is $0.05/30 = 0.00167$, and $z = 2.94$ and $z = 3.14$ are the normal quantiles named above. Holm compares 0.0083 with 0.00167 and stops. No rate is outside the corrected rules.

A binomial test of three misses in thirty trials at $p = 0.05$ has upper tail 0.19. The expected count under independence is 1.5. The same within-cell dependence applies, so 0.19 is the independent-trials figure. On that calculation, three misses is within the count chance predicts. Three rates each miss their own 5% band, and three misses in thirty trials is an ordinary count for that band.

The unbalanced power rates are 49.35, 21.28, and 8.51 against 49.23, 21.08, and 7.77. The largest $|z|$ in that cell is 1.85, on the third rate. The averages are $-0.403$, $-0.215$, and 0.0050 against $-0.404$, $-0.218$, and 0.0054. Convergence there is 8,982 of 10,000, 89.82%, against 89.29%.

Twenty-nine of the 30 averages are inside two combined standard errors. The exception is the multigroup equal-group null contrast 3: 0.0023 against $-0.0021$, $|z| = 2.40$. The rejection rate in that cell is 5.27 against 5.39. The 30 averages were not the family that Bonferroni and Holm were applied to. The same $z = 2.94$ threshold would place this one average inside a corrected band.

Table 4.5's multigroup null row at $N = 240$, condition 3, prints 9,675 and 95.75%. Those two figures are not the same rate: 9,675/10,000 is 96.75. Our count is 9,656 of 10,000, 96.56%. That rate is inside two combined standard errors of 96.75 and outside two combined standard errors of 95.75 ($|z| = 2.98$ against the printed percent). The MIMIC true-difference rows print 10,000 of 10,000. Ours are 9,988 and 9,986. The plug-in standard error gives a printed rate of 100% no

variance, so those gaps exceed two combined standard errors. The rejection rates in those cells do not.

### 3.6 The named installation

The question is whether those gaps are the R installation. The dissertation names R 4.2.2, simsem 0.5.16, and lavaan 0.6.17, and it names `nlminb` with 10,000 iterations. It does not name an operating system, and it does not set `RNGkind`. The default in R since 3.6.0, and the default in the R 4.3.3 used here, is Mersenne-Twister, Inversion, and Rejection. `sim()` in simsem 0.5-16 replaces that generator with L'Ecuyer-CMRG and then applies the seed passed to `exportData`.

The OSF node has no `sessionInfo`, no log, and no `.Rhistory`. The files were uploaded on 2025-09-20. The generation scripts call `set.seed(1313)` and pass seed 13131 to `exportData`. That call is commented out. `exportData(program = "Mplus")` writes `.dat` files. The estimation scripts call `lavaan::sem`.

simsem 0.5-16 was the CRAN release from 2021-03-28 until 0.5-17 on 2025-04-02, after the November 2024 dissertation. For a lavaan script, its generator is `lavaan::simulateData`. The 0.5-17 change in that function is a class check. lavaan 0.6-17, the named version and the version used here, was published on 2023-12-20. R 4.2.2 was not installed. The redraw uses simsem 0.5-16, lavaan 0.6-17, and `sim(dataOnly = TRUE, seed = 13131)`. The script is `simulation/env_miss.R`. The summary is `simulation/out/env_miss.csv`.

On that stream the three MIMIC rates are 22.53 against 23.37 ($|z| = 1.42$), 6.46 against 7.41 ($|z| = 2.64$), and 4.86 against 5.32 ($|z| = 1.49$). Two fall inside the uncorrected band. The third, MIMIC with shares 45/45/5/5, null means, contrast 2, is the same 6.46 the hand-written generator produced. The lavaan converged flag and the finite standard-error rule selected the same replicates in every cell of this redraw. No captured warning contained the word "gradient." Requiring nonnegative variances dropped replicates only in the multigroup cell, 9,774 of 9,927, and that subset does not move the rates onto the page.

The multigroup equal-group null contrast 3 has average 0.0015 against −0.0021. Taking the printed standard error 0.0013 as the replicate standard deviation, $|z|$ is 1.98. The nonnegative-variance subset has average 0.0014, $|z| = 1.92$. The earlier stream's average was 0.0023, $|z| = 2.40$. The gap sits on the boundary of the uncorrected band. It is inside $z = 2.94$. Convergence in that cell is 9,927 of 10,000, against Table 4.5's 9,932 (99.32%).

The MIMIC rows of Table 4.5 print 10,000 of 10,000. This stream gives 9,988 and 9,980 in the two true-difference cells, and 9,964 and 9,951 in the two null cells. The finite-standard-error rule matches the converged flag, so the shortfall is not a stricter filter.

Mplus was not the estimator. The page quotes a lavaan warning and names `nlminb`. The OSF estimation scripts call `lavaan::sem`. No Mplus output is on the node. Had Mplus been the estimator, its default would fix the first loading at 1, and a default summary prints a convergence criterion of 0.00005 with at most 1,000 iterations. A negative residual variance is reported and the solution is still returned. Those defaults are not the path that produced the tables.

Public code search did not turn up the dissertation scripts. The record is `simulation/out/code_search.md`. The only related public file is https://github.com/jjeffries13/APA_2024, a ten-indicator simulation that reuses the loading idiom and seed 13131. It is not this design.

`Dynamic-Fit-Indices/DFI_Presentation.Rmd` calls `sessionInfo()` and sinks the text to `DFIsessionInfo.txt`. That text file is not in the repository. The commits run from March through October 2022, the file is dated 9 March 2022, and the working directory is under `/Users/jayjeffries/`. The slides beside it are a Google deck and contain no session block. Two teaching repositories last changed in February and June 2023 carry a badge for R 4.2.1. A May

2022 file has a commented Windows path under a UNL OneDrive account. The paths that the same period actually reads sit on that Mac home directory. No repository from 2023 or 2024 contains a saved `sessionInfo`, an `renv.lock`, a packrat file, or a package `DESCRIPTION`. The dissertation text remains the source for R 4.2.2. A Mac path in other projects does not name the machine that produced the 10,000-replicate tables, and it supplies no lavaan or simsem version. A BLAS difference would move a converged estimate in the last digits. It would not move a rejection rate by a percentage point. Those cells were not rerun.

# 4 Assumptions

## 4.1 Random-number stream

Jeffries sets `set.seed(1313)` and passes seed 13131 to `exportData`. The 400-replicate screen used the L'Ecuyer generator, `set.seed(13131)`, and one drawn seed per cell. The 10,000-replicate run sets a separate fixed seed in each cell, listed in `simulation/out/seeds_10k.csv`. The replicates are not Jeffries's replicates. The `.dat` files are not in the OSF archives.

## 4.2 Package versions

The dissertation names R 4.2.2 and simsem 0.5.16. The ten-cell Monte Carlo used R 4.3.3 and simsem 0.5.17. The miss-cell redraw installed simsem 0.5-16 and kept lavaan 0.6.17. R 4.2.2 was not installed. The optimizer is `nlminb`. A replicate counts as converged when `lavInspect(fit, "converged")` is true, which is the flag the OSF script reads from the summary header, and the three contrast standard errors are finite. In the simsem 0.5-16 redraw those two rules selected the same replicates.

## 4.3 Replicate count

The tables use 10,000 converged replicates. The first classification uses 400 attempts per cell. That length separates a rate of 7.8% from a rate of 4.5%, and it leaves a wide standard error on our side of each comparison. The classification at the printed length uses 10,000 attempts in each of the ten cells. The seeds are fixed and are listed in `simulation/out/seeds_10k.csv`.

## 4.4 The first intercept

The PDF page writes `i1 ~ 1*`. The OSF generation script writes `i1 ~ 1*1`, as do the other three indicators and the null model on the page. We generated from `1*1`.

# 5 Ambiguities that were resolved

## 5.1 The ink matches the text layer

The question was whether a sign or a printed average in the text extract disagreed with the page. On the pages checked here, it does not. Figure 2 has the minus signs on −0.40, −0.20, and 0.

The crops are `paper/figures/`. The script is `paper/figures/crop_pages.py`. It runs `pdftoppm -png -r 600`. Pages whose rotation flag is 90 degrees are turned clockwise into reading order. The command is `python3formalization/latent-contrasts/paper/figures/crop_pages.py`.

Latent variables were identified using the marker method approach, where the first indicator's loading was fixed to one ($\lambda_1 = 1$). Per recommendations from Hancock (1997), over both SEM forms, intercepts were modeled such that $\tau_{1_1} = \tau_{2_1} = \ldots \tau_{x_i}$. Further, the variance of all latent variables were freely estimated.

Figure 1: Marker identification. The first loading is fixed at 1, and the intercepts are equated. PDF page 90.

model results for models of data that were explicitly generated to show latent mean differences (see Tables 4.1 and 4.2). As a reminder, the data-generated latent mean difference for Contrasts 1 through 3 were, on average, -0.40, -0.20, and 0 (i.e., a null difference), respectively. In the null model, the data-generated mean differences for Contrasts 1 through 3 were zero, on average, across all three contrasts. An average

Figure 2: The data-generating contrasts are $-0.40$, $-0.20$, and 0. PDF page 116.

### 5.2 The public analysis script

The question was which identification produced Tables 4.1 and 4.3. The OSF project https://osf.io/q9yu8/ contains the multigroup and MIMIC scripts and the summary files. The multigroup summary matches Table 4.1 to the printed digits, including 8,723 significant replicates of contrast 1 at $N = 240$ in the equal true-difference cell. The standard-error column in that file is the standard deviation divided by $\sqrt{10{,}000}$.

The artifact is `osf/analysis/mgcfa_model.R`, retrieved 2026-10-27. The moment fit of that string on the parsed generation population is the fifth row of Table 2. The Monte Carlo is `simulation/out/validate_gaps.csv`, from `Rscriptvalidate_gaps.R400out`.

### 5.3 The third code and the numeric codes

Appendix A and the OSF generation script code `c3` with the opposite sign from Table 2.2. On the printed means the third contrast is zero, so the sign does not move a two-sided rejection rate for that contrast. Numeric codes $\pm 1$ estimate half the defined contrast, because the code changes by 2. Factor codes with a free latent intercept estimate the full contrast. Theorem 159 is the 0/1 form of that identity. The marker theorem at 131 is the every-loading-0.70 reading, and it is not the parsed generation population.

## 6 What remains unresolved

One MIMIC rate stays outside two combined standard errors after the simsem 0.5-16 redraw: shares 45/45/5/5, null means, contrast 2, 6.46 against 7.41. Both generators produced that 6.46. The rate is inside Bonferroni and Holm on the family of thirty. The multigroup equal-group null contrast 3 has average 0.0015 against $-0.0021$, on the boundary of the uncorrected band. The

```
# ---- Models of N = 240 ----
pacman::p_load(here, tidyverse, simsem, lavaan, purrr)
set.seed(1313)
# ---- Significance Model ----
## ---- Model Syntax ----
# 4 item model
pop_model_cond1 <- "
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~~ 0.51*i1
i1 ~ 1*
i2 ~~ 0.51*i2
i2 ~ 1*1
i3 ~~ 0.51*i3
```

Figure 3: Appendix A. The first intercept is written `1*`. The loading line contains both `NA*i1` and `0.7*i1`.

MIMIC convergence counts remain short of the printed 10,000 of 10,000. The `.dat` files are not on OSF, so this is not Jeffries's stream. R 4.2.2 was not the installed binary.

The page prints $-0.4001$ for the MIMIC average of contrast 1 at $N = 240$ in the equal true-difference cell. The public summary stores $-0.399094$. The significance count in that cell is 9,573 on the page and in the file. The other MIMIC cells we checked round from the file onto the page.

The PDF generation line `i1 ~ 1*` does not parse. The OSF script's line `i1 ~ 1*1` does. One uncommented `exportData` call, in the $N = 480$ condition-3 null script, names group sizes $(168, 168, 72, 72)$, which is the 35/15 split. Some commented calls name sizes from a different condition than the file name. The datasets those calls would have written are not in the archive. The Monte Carlo in Table 3 uses the printed shares, and the condition-3 averages and the 89.29% convergence rate agree with the page.

Jeffries and Bovaird (2026), DOI `10.1177/01466216261452255`, was not obtained as a PDF. Unpaywall, Europe PMC, and the Wayback machine had no PDF on 2026-10-27. The OSF node is the analysis record for that title. It does not contain the article.

## 7 Code and source availability

| Place | Date | Result |
|---|---|---|
| Appendix A of the dissertation PDF | 2026-10-27 | Generation syntax. sha256 of the PDF is `a21df98c...fd469b3` |
| OSF `/v2/search/` for the title | 2026-10-27 | HTTP 404 |
| OSF node `q9yu8` by title filter | 2026-10-27 | Public project, analysis scripts and summaries, no `.dat` files and no article PDF |
| DOI `10.1177/01466216261452255` | 2026-10-27 | No PDF from the publisher page, Unpaywall, Europe PMC, or Wayback |
| lavaan 0.6.17 | 2026-10-27 | Installed and used for every fit |
| simsem 0.5.17 | 2026-10-27 | Installed. Rejects `i1 ~ 1*`. |
| simsem 0.5-16 from the CRAN archive | 2026-10-28 | Installed for `env_miss.R`. Generator is `lavaan::simulateData`. |
| GitHub, GitLab, Zenodo code search | 2026-10-28 | No dissertation script. Record in `simulation/out/code_search.md`. |
| OpenMx, MplusAutomation, Mplus | 2026-10-27 | Not installed |

The dissertation file was retrieved from the Wayback `id_` object dated 2025-05-11, snapshot `20250511025154`. The help request for a script, filed before this OSF node was found, is https://github.com/alectionary/vibe-math/issues/33. The fetch request for the 2026 article is https://github.com/alectionary/vibe-math/issues/34.

## 8 Next steps

The rate that both generators put at 6.46 against 7.41, the average on the boundary of the uncorrected band, and the MIMIC convergence counts short of 10,000 are the gaps the named simsem release did not close. The missing `.dat` files would be Jeffries's own stream. An R 4.2.2 binary would test whether L'Ecuyer draws differ from R 4.3.3. The 2026 PDF would show whether the journal version uses these tables.

## 9 What we proved

The codes, in group order $(1, 2, 3, 4)$, are

$$c_1 = (-1, 1, -1, 1), \qquad c_2 = (-1, -1, 1, 1), \qquad c_3 = (1, -1, -1, 1).$$

The weighted product is $\sum_{g=1}^{4} n_g c_{jg} c_{kg}$. For equal sizes the three products are zero (107). For sizes $(a, a, b, b)$ the products of $c_1$ with $c_2$ and of $c_2$ with $c_3$ stay zero, and the product of $c_1$ with $c_3$ equals $2(b - a)$ (68). The column sums are zero (99).

A defined contrast is the average of the means coded $+1$, minus the average of the means coded $-1$. On $\kappa = (1, 3/5, 4/5, 2/5)$ the three contrasts are $-2/5$, $-1/5$, and 0 (123). The third is zero while $\kappa_1 \neq \kappa_2$ (127). Multiplying every mean by $7/10$ multiplies the contrasts by $7/10$ (131). That factor is the every-loading-0.70 marker reading. It is not the population lavaan builds from `NA*i1`. The treatment-contrast form with coefficients $-2/5$, $-1/5$, and 0 reproduces $\kappa$ (159).

The proofs are rational arithmetic. The `ring` proofs depend on `propext`, `Classical.choice`, and `Quot.sound`. The `native_decide` proofs also depend on `Lean.ofReduceBool`. There is no `sorry`.

## 10 Artifacts

- Code identities. File `LatentContrasts/Orthogonality.lean`. Command: `lakebuild`.
- Moment fits of every reading. File `simulation/out/candidate_readings.csv`. Command: `Rscriptcandidate_readings.Rout`. Seed 13131 for the simulated samples. Moment fits do not use a seed.
- Large-sample MIMIC. File `simulation/out/mimic_large.csv`. Command: `Rscriptmimic_large.Rout`. Seed 7.
- Monte Carlo. File `simulation/out/validate_gaps.csv`. Command: `Rscriptvalidate_gaps.R400out`.
- Agreement classification, 400 replicates. File `simulation/out/gap_scores.csv`, with the family summary in `simulation/out/gap_scores_multiplicity.csv`. Command: `Rscriptscore_gaps.Rout/validate_gaps.csvout/gap_scores.csv`.
- Ten-thousand-replicate rerun. File `simulation/out/validate_10k.csv`, with one checkpoint per cell in `simulation/out/cell10k_*.csv`. Seeds: `simulation/out/seeds_10k.csv`. Command: `Rscriptvalidate_10k.R10000out`.
- Agreement classification at that length. File `simulation/out/gap_scores_10k.csv`, with the family summary in `simulation/out/gap_scores_10k_multiplicity.csv`. Command: `Rscriptscore_gaps.Rout/validate_10k.csvout/gap_scores_10k.csv`.
- simsem 0.5-16 redraw of the miss cells. File `simulation/out/env_miss.csv`. Command: `Rscriptenv_miss.R10000out`, with simsem 0.5-16 on the library path. Versions: `simulation/out/env_session.txt`.
- Code search. File `simulation/out/code_search.md`.
- Equal-loading replay, kept so the old averages stay reproducible. File `simulation/out/monte_carlo.csv`. Command: `Rscriptmonte_carlo.R200100out`.
- Parser checks. File `simulation/out/parser_checks.csv`.
- Session. File `simulation/out/session_info.txt`. Command: `Rscriptsession_info.Rout`.
- OSF scripts and summaries. Directory `osf/`.
- Page crops. Directory `paper/figures/`.

The dissertation PDF is local and gitignored. Its sha256 is `a21df98c20401491e6b4a1d00a21b6f7cee5551a00c04a89fb414ec6cfd469b3`.

**Table 4.5**
*Initial Model Convergence Rates by Condition & Model*

| *N* | Condition | MIMIC Frequency Converged (%) | MGCFA Frequency Converged (%) |
|---|---|---|---|
| Significance Models | | | |
| 240 | 1 | 10000 (100%) | 9886 (98.86%) |
| | 2 | 10000 (100%) | 9811 (98.11%) |
| | 3 | 10000 (100%) | 8929 (89.29%) |
| 480 | 1 | 10000 (100%) | 9989 (99.89%) |
| | 2 | 10000 (100%) | 9994 (99.4%) |
| | 3 | 10000 (100%) | 9805 (98.05%) |
| 960 | 1 | 10000 (100%) | 10000 (100%) |
| | 2 | 10000 (100%) | 10000 (100%) |
| | 3 | 10000 (100%) | 9986 (99.86%) |
| Null Models | | | |
| 240 | 1 | 10000 (100%) | 9932 (99.32%) |
| | 2 | 10000 (100%) | 9911 (99.11%) |
| | 3 | 10000 (100%) | 9675 (95.75%) |
| 480 | 1 | 10000 (100%) | 9998 (99.98%) |
| | 2 | 10000 (100%) | 10000 (100%) |
| | 3 | 10000 (100%) | 9999 (99.99%) |
| 960 | 1 | 10000 (100%) | 10000 (100%) |
| | 2 | 10000 (100%) | 10000 (100%) |
| | 3 | 10000 (100%) | 10000 (100%) |

Figure 4: Table 4.5. At $N = 240$ and condition 3 the multigroup significance model converged in 8,929 of 10,000 replicates.